

Vivek Singh

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Career Objective

Computer Science undergraduate with strong foundation in Software Testing, Database Validation, and Debugging, seeking a Graduate Engineer Trainee role at Wipro. Skilled in preparing test data, executing unit tests, performing database checks, and documenting results as per defined guidelines, with working knowledge of Java, Python, SQL, and Spring Boot. Quick learner, eager to build domain expertise while contributing to reliable, well-tested software delivery.

Education

Maharana Pratap Engineering College

Kanpur, Uttar Pradesh

Bachelor of Technology in Computer Science and Engineering

2023 – 2027

– CGPA: 7.5/10

– Relevant Coursework: Data Structures, Algorithms, Software Engineering, Operating Systems, Computer Networks, Database Management Systems, Theory of Automata

Technical Skills

- **Languages:** Java, Python, C++, SQL
- **Frameworks & Libraries:** Spring Boot, OpenCV, dlib
- **Databases:** MySQL, MongoDB
- **Tools:** Git, GitHub, IntelliJ IDEA, VS Code
- **Software Testing:** Test Data Preparation, Test Case Execution, Unit Testing, Database Validation, Verification, Debugging, Defect Identification, Test Documentation, Result Analysis
- **Core Concepts:** Data Structures & Algorithms, OOP, DBMS, SDLC, REST APIs, Quality Assurance
- **Soft Skills:** Analytical Thinking, Problem Solving, Attention to Detail, Communication, Team Collaboration

Projects

Facial Recognition Attendance System

2025

Python, OpenCV, dlib, MongoDB

- Developed a real-time facial recognition attendance system using Python, OpenCV, and MongoDB with more than 95% face detection accuracy.
- Prepared initial test data and performed unit testing and validation of face recognition modules under different lighting conditions to verify expected vs. actual results.
- Executed database checks on MongoDB attendance records, verifying data consistency and integrity as per defined validation criteria.
- Debugged recognition failures, analyzed root causes, and documented test outcomes and performance results in prescribed formats.
- Implemented liveness detection to prevent spoofing attacks and improve overall system security.

AI Image Screenshot Protection Model

2026

Python, OpenCV

- Designed an AI-based image protection system capable of identifying unauthorized screenshots in real time.
- Prepared test data and executed unit testing across multiple scenarios, verifying screenshot detection accuracy against expected results.
- Performed debugging and optimization to reduce inference latency below 100 ms while maintaining high detection accuracy.
- Documented test results, analyzed performance metrics, and maintained test records for continuous improvement.
- Applied image preprocessing and computer vision techniques to enhance model reliability and robustness.

2D Physics Simulator

2023

Python

- Developed a real-time 2D physics simulator implementing gravity, motion, and collision mechanics using object-oriented programming principles.
- Performed functional testing and validation of physics calculations, verifying simulation outputs against expected results under different scenarios.

Achievements

- Solved 200+ Data Structures and Algorithms problems on LeetCode covering Arrays, Strings, Trees, Graphs, Dynamic Programming, Greedy Algorithms, and Binary Search.
- Achieved a 5-Star Java Rating on HackerRank demonstrating strong programming and problem-solving skills.
- Developed multiple academic projects following SDLC practices including coding, unit testing, debugging, database validation, and technical documentation.
- Strong understanding of Software Testing fundamentals including Test Data Preparation, Test Case Execution, Database Validation, and Defect Identification.

Certifications

- IBM Data Visualization using Python
- Oracle Cloud Infrastructure 2025 Generative AI Professional

Additional Information

- Comfortable working with Java, SQL, MySQL, MongoDB, Spring Boot, Git, and REST APIs.
- Familiar with SDLC, Software Testing, Unit Testing, Debugging, Test Case Execution, Database Validation, and Technical Documentation.
- Strong analytical, communication, problem-solving, and collaboration skills with a keen attention to detail.
- Actively learning Backend Development, Software Quality Engineering, Modern Testing Practices, and open to building domain knowledge in Health and Welfare (HW) Tech.